

**SIMATS ENGINEERING**  
**Department of Computer Science & Engineering**  
**ASSESSMENT TOOL 1- ANALYTICAL PROBLEM SOLVING**

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|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------------------------------|
| <b>Course Code:</b>   | <b>CSA1209</b>                                                                                                                                                                                                                    | <b>Course Title:</b> | <b>Computer Architecture</b>                                |
| <b>CO Assessed:</b>   | <b>CO2</b>                                                                                                                                                                                                                        | <b>Assessment:</b>   | <b>ASSESSMENT TOOL1-<br/>ANALYTICAL PROBLEM<br/>SOLVING</b> |
| <b>Weightage:</b>     | <b>45%</b>                                                                                                                                                                                                                        | <b>Total Marks:</b>  | <b>25</b>                                                   |
| <b>CO2 Statement:</b> | Apply integer and floating-point arithmetic algorithms including Booth's multiplication, restoring/non-restoring division, and IEEE 754 standard operations to solve fixed-point and floating-point computational problems. (BL3) |                      |                                                             |
| <b>Student Name:</b>  |                                                                                                                                                                                                                                   | <b>Reg.No.:</b>      |                                                             |
| <b>Department:</b>    |                                                                                                                                                                                                                                   | <b>Date:</b>         |                                                             |

**ANNEXURE A**

1. A processor performs arithmetic operations on signed 8-bit integers using 2's complement representation. Consider two numbers: +45 and -23. Analyze in detail how the system performs both addition and subtraction, including binary conversion, alignment of sign bits, and intermediate steps. Determine whether overflow occurs, and explain how the processor detects overflow and interprets the final result.
2. A high-performance computing system replaces a ripple carry adder with a 4-bit carry look-ahead adder (CLA) to reduce computation delay. Given two binary inputs, analyze how generate and propagate signals are formed and how carries are computed in advance. Compare the time delay of CLA with ripple carry addition and justify, with reasoning, why CLA improves speed in modern processors.
3. In a signed multiplication operation, a processor uses Booth's algorithm to multiply two numbers, -6 and +5. Analyze the algorithm step-by-step, including the role of the accumulator, multiplier, and Booth bit, along with arithmetic shifts. Explain how the algorithm efficiently handles signed numbers and reduces the number of addition/subtraction operations compared to traditional multiplication.
4. A processor is required to perform division of two binary numbers (e.g.,  $13 \div 3$ ) using both restoring and non-restoring division algorithms. Analyze both methods step-by-step, showing intermediate remainders and quotient formation. Compare the two approaches in terms of number of steps, complexity, and efficiency, and explain why non-restoring division is often preferred in modern systems.
5. A scientific application requires precise handling of real numbers using the IEEE 754 standard for floating-point representation. Given a decimal number (e.g., -13.25), analyze how it is represented in both single precision and double precision formats, including sign, exponent, and mantissa fields. Further, explain how floating-point addition is performed between two such numbers, highlighting issues like normalization, rounding, and precision loss.

**Code of Conduct:**

I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:**

**MARKS ALLOCATION**

|           | <b>Problem Understanding<br/>(20%)</b> | <b>Method Selection<br/>(20%)</b> | <b>Solution Process<br/>(25%)</b> | <b>Accuracy of Calculation<br/>(20%)</b> | <b>Interpretation of Results<br/>(15%)</b> |
|-----------|----------------------------------------|-----------------------------------|-----------------------------------|------------------------------------------|--------------------------------------------|
| <b>Q1</b> |                                        |                                   |                                   |                                          |                                            |
| <b>Q2</b> |                                        |                                   |                                   |                                          |                                            |
| <b>Q3</b> |                                        |                                   |                                   |                                          |                                            |
| <b>Q4</b> |                                        |                                   |                                   |                                          |                                            |
| <b>Q5</b> |                                        |                                   |                                   |                                          |                                            |

**RUBRICS FOR CALCULATION:**

| <b>Criteria</b>           | <b>Weightage</b> | <b>Level 4 – Exemplary</b>                                                        | <b>Level 3 – Proficient</b>               | <b>Level 2 – Developing</b>                | <b>Level 1 – Beginning</b>                    |
|---------------------------|------------------|-----------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------|-----------------------------------------------|
| Problem Understanding     | 20%              | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps       | Partial understanding; misses key elements | Misinterprets or unable to understand problem |
| Method Selection          | 20%              | Selects most appropriate method/formula with strong justification                 | Selects correct method with minor errors  | Inappropriate or partially correct method  | Unable to select method                       |
| Solution Process          | 25%              | Logical, systematic steps with correct approach throughout                        | Mostly correct steps with minor mistakes  | Disorganized steps; multiple errors        | No clear process                              |
| Accuracy of Calculation   | 20%              | All calculations accurate; correct final answer                                   | Minor calculation errors                  | Frequent errors; incorrect final answer    | No valid calculations                         |
| Interpretation of Results | 15%              | Clearly interprets results with units and engineering relevance                   | Basic interpretation with minor omissions | Limited or unclear interpretation          | No interpretation                             |

## CO2 ATI

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Course: Computer Architecture -  
CSA1804

### 1. 2's Complement Addition and Subtraction

Given:

$$A = +45$$

$$B = -23$$

Binary Conversion

$$+45 = 00101101$$

$$+23 = 00010111$$

$$-23 = 11101001 \text{ (2's Complement)}$$

Addition:  $45 + (-23)$

$$\begin{array}{r} 00101101 \\ + 11101001 \\ \hline 100010110 \end{array}$$

Ignore the carry:  $00010110 = 22$

$$\therefore 45 + (-23) = 22$$

Subtraction:  $45 - (-23)$

Subtraction is addition of 2's complement:

$$45 - (-23) = 45 + 23$$

$$\begin{array}{r} 00101101 \\ + 00010111 \\ \hline 01000100 \end{array}$$

$$01000100 = 68$$

$$45 - (-23) = 68$$

Overflow: No Overflow occurs because the sign of the result is valid. In



## 2. 4-bit Carry Look-Ahead Adder

For each bit:

$$P_i = A_i \oplus B_i = A_i B_i$$

Carry:

$$C_{i+1} = G_i + P_i C_i$$

For example:

$$A = 1011$$

$$B = 0110$$

CLA calculates all carries directly:

$$C_1 = G_0 + P_0 C_0 \quad C_2 = G_1 + P_1 G_0 + P_1 P_0 C_0$$

and similarly for  $C_3, C_4$

Advantage

Ripple Carry Adder:

$$FA \rightarrow FA \rightarrow FA \rightarrow FA$$

Carry must pass through each stage.

CLA:

$$P, G \rightarrow \text{Carry Logic} \rightarrow \text{All Carries}$$

Carries are calculated in parallel, so CLA is faster, although it requires more hardware.

## 3. Booth's Algorithm: $-6 \times 5$

Given:

$$M = -6 = 1010$$

$$Q = +5 = 0101$$

$$A = 0000$$

$$Q_{-1} = 0$$

Booth rules:

| $Q_0$ | $Q_{-1}$ | Operation    |
|-------|----------|--------------|
| 0     | 0        | No Operation |
| 0     | 1        | $A = A + M$  |
| 1     | 0        | $A = A - M$  |
| 1     | 1        |              |

Perform the operation and arithmetic right shift for 4 cycles.

Final result

$$AQ = 11100010$$

$$11100010 \text{ in 2's complement} = -30.$$

$$(-6) \times 5 = -30$$

Advantage: Booth's algorithm efficiently handles signed numbers and reduces the number of additions/subtractions, especially for consecutive 1s.

#### 4. Restoring and Non-Restoring Division

Given:

$$13 \div 3$$

Binary:

$$13 = 1101$$

$$3 = 0011$$



Expected:

$$13 \div 3 = 4 \text{ remainder } 1$$

### Restoring Division

Basic steps:

1. Shift remainder and quotient left.
2. Subtract divisor.
3. If remainder becomes negative, restore it by adding divisor.
4. Set quotient bit accordingly.

Final:

$$\text{Quotient} = 0100$$

$$\text{Remainder} = 0001$$

### Non-Restoring Division

Instead of immediately restoring a negative remainder;

- \* If remainder is positive  $\rightarrow$  subtract divisor.
- \* If remainder is negative  $\rightarrow$  add divisor.
- \* Correct the final remainder if necessary.

Final:

$$\text{Quotient} = 0100$$

$$\text{Remainder} = 0001$$

## Comparison

### Restoring

Restores negative remainder

More operations

Slower

Simpler

### Non-Restoring

Does not immediately restore

Fewer operations

Faster

More efficient

$\therefore$  non-restoring division is generally preferred for high-speed implementations.

## 5. IEEE 754 Representation of -13.25

Convert to binary:

$$13.25 = 1101.01_2$$

Normalize:

$$1101.01 = 1.10101 \times 2^3$$

Single Precision

\* Sign = 1

\* Exponent =  $3 + 127 = 130 = 10000010$

\* Fraction = 101010000000000000000000

Therefore:

1 10000010 101010000000000000000000



